

Sarfraj Ahmad Raza

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Technical Skills

Languages: JavaScript, TypeScript, Python, HTML, CSS

Frontend: React.js, Next.js, Redux Toolkit, RTK Query, Material UI, Chakra UI, React Router, Formik, Axios, Chart.js

Backend & Database: Node.js, Express.js, FastAPI, REST APIs, MongoDB, Mongoose

AI / LLM: LangChain, LangGraph, RAG, MCP, LLM Tool Calling, Human-in-the-Loop (HITL), Prompt Engineering

AI APIs & Concepts: OpenAI API, OpenRouter, Hugging Face, Tavily, Embeddings, Vector Search, Semantic Retrieval

Developer Tools: Git, GitHub, VS Code, Postman, Swagger, Jira, ESLint, Prettier

Deployment & Performance: GitHub Actions, Vercel, Netlify, Lighthouse, Lazy Loading, Code Splitting

Experience | SOFTWARE ENGINEER (JUL-2023 - Till Date)

Quicktouch Technologies Pvt. Ltd.

Software Engineer

Jul 2023 – Present

NSP, New Delhi, Delhi

- **Engineered** ERP systems for schools and colleges, building robust modules for **students, faculty, and administrators** using **React.js** and **Material-UI**, significantly enhancing operational efficiency across institutions.
- **Developed** end-to-end modules including **Result, Fee, Visitor Management, and Attendance**, transforming manual processes into fully digital workflows and improving **data accuracy** across schools and colleges.
- **Empowered 3,500+ schools** with automated templates for **result cards, fee slips, payslips, salary bills, and certifications**, drastically reducing manual processing time and administrative overhead.
- **Implemented Import/Export** features for bulk data handling, simplifying **data migration** tasks and streamlining document processing across large institutions.
- **Optimized** API integrations using **Redux Toolkit (RTK) Queries** with enhanced **caching, lazy loading, and data synchronization**, delivering significant frontend performance improvements.
- **Resolved** real-time **production bugs** with quick turnaround times, ensuring **uninterrupted platform operations** and maintaining high availability for **100,000+** daily active users.
- **Improved** application performance by implementing **code splitting, lazy loading, and code optimization**, significantly reducing load times and enhancing overall user experience.

Projects

QuickCampus ERP | *React.js, Node.js, Material UI, Redux Toolkit Query, JavaScript*

- Contributed to the development and scaling of a cloud-based school management ERP serving **3,500+ schools** and **100,000+ daily active users**, building high-performance React.js interfaces and reusable UI component libraries.
- Designed and deployed an end-to-end attendance management system including real-time attendance tracking, GPS-based geofencing validation, work-from-home logging, leave management, hierarchical approval workflows, and automated email notifications – eliminating reliance on third-party tools.
- Integrated **OpenAI API** into the ERP platform, enabling AI-powered content generation, automated administrative responses, and smart data insights, reducing manual workload for school administrators.
- Resolved real-time production bugs across multiple modules, improving platform stability and ensuring uninterrupted service for **100,000+** daily active users.

- Engineered a Node.js PDF generation service that takes HTML templates and dynamic JSON data as input and returns server-rendered PDFs as output, eliminating cross-device rendering inconsistencies and ensuring pixel-perfect document consistency across all platforms.

AI Executive Assistant | Python, LangGraph, LangChain, MCP, FastAPI, React.js, OpenAI

[GitHub](#) 

- Engineered an **agentic AI executive assistant** using LangGraph and MCP capable of autonomously reasoning, selecting tools, executing multi-step workflows, observing results, and dynamically determining subsequent actions.
- Integrated **Gmail, Google Calendar, Google Meet, and real-time web search**, enabling the agent to search and manage emails, create/update/delete events, schedule meetings, perform research, and execute independent tools in parallel.
- Implemented **Human-in-the-Loop (HITL)** approval for sensitive actions and automated PDF/Excel generation, enabling workflows such as research → document generation → approval → email delivery with real file attachments.

YouTube RAG Assistant | Python, LangChain, RAG, Vector Database, OpenAI, YouTube Transcripts

[GitHub](#) 

- Built a **Retrieval-Augmented Generation (RAG)** application that processes YouTube video transcripts and enables users to ask natural-language questions and receive context-aware answers grounded in video content.
- Implemented an end-to-end RAG pipeline for **transcript extraction, text chunking, embedding generation, vector storage, semantic retrieval, and LLM-based response generation**.
- Optimized retrieval by passing only the most relevant transcript chunks to the LLM, reducing unnecessary context and **token consumption** while improving answer relevance.

NDA Securities | Next.js, Nest.js, TypeScript, REST APIs

[Link](#) 

- Led frontend development of a fintech-grade stock market web application, integrating **real-time stock market APIs** to deliver live price feeds, market data visualization, and dynamic financial dashboards.
- Built a fully dynamic **CMS admin panel** enabling non-technical administrators to manage and update all platform content independently, eliminating engineering dependency for routine content operations.
- Delivered the project to enterprise-level standards under strict NDA, ensuring high performance, security, and reliability across all user-facing and admin interfaces.

JSON Viewer | React.js, Chakra-UI, CSS

[Link](#) 

- Developed a JSON Viewer application that allows users to load data from APIs, format JSON, remove white spaces, and parse data efficiently.
- Implemented a tree-view format for displaying parsed JSON data, improving readability and ease of navigation.
- Enhanced user experience by adding formatting and white space removal features.

Education

Guru Gobind Singh Indraprastha University
Bachelor of Technology in Information Technology

New Delhi

Place: New Delhi

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